Bozheng Li

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Research Interest

My research centers on **multimodal video understanding**, from deploying models in product-level applications to enabling real-world spatial intelligence, with the long-term goal of building **proactive video agents** that can perceive, reason, and act across both the digital and physical worlds.

Education

Brown University **09/2024 – 05/2026(Expected)**
Sc.M in Computer Science *Providence, RI*

- **GPA:** 4.00/4.00
- **Core Modules:** Deep Learning, Advanced Deep Learning, Reintegrated AI, Self-Supervised Learning

Zhejiang University **09/2020 – 06/2024**
B.Eng in Information Engineering *Hangzhou, China*

- **GPA:** 3.97/4.00, Rank: 7/142
- **Core Modules:** Artificial Intelligence, Computer Vision

Selected Publications

Bozheng Li, Wanhua Li, Hanspeter Pfister. PVD: Probing 3D awareness of VideoLLM. In: *submission to CVPR 2026.*

Bozheng Li, Yongliang Wu, Yi Lu, Jiashuo Yu, Licheng Tang, Jiawang Cao, Wenqing Zhu, Yuyang Sun, Jay Wu, Wenbo Zhu. VEU-Bench: Towards Comprehensive Understanding of Video Editing. In: *Proceedings of the Computer Vision and Pattern Recognition Conference (**CVPR**)*. 2025.

Bozheng Li, Mushui Liu, Gaoang Wang, Yunlong Yu. Frame Order Matters: A Temporal Sequence-aware Model For Few-shot Action Recognition. In: *Proceedings of the AAAI Conference on Artificial Intelligence (**AAAI**)*. 2025.

Yongliang Wu, **Bozheng Li**, Jiawang Cao, Wenbo Zhu, Yi Lu, Weiheng Chi, Chuyun Xie, Haolin Zheng, Ziyue Su, Jay Wu, Xu Yang. Zero-shot Long-form Video Understanding Through Screenplay. In *arxiv.* 2024

Mushui Liu, **Bozheng Li**, Yunlong Yu. OmniCLIP: Adapting CLIP for Video Recognition with Spatial-Temporal Omni-Scale Feature Learning. In: *Proceedings of European Conference on Artificial Intelligence (**ECAI**)*. 2024.

Research Experience

PVD: probing 3D awareness of VideoLLM. **06/2025 – 11/2025**
Harvard University *Research Assistant*

- Diagnosed overfitting of current 3D-aware VideoLLMs to flawed benchmarks and their poor generalization in spatial understanding beyond biased training data, indicating a lack of robust spatial understanding capabilities.
- Probed 3D awareness rise from the pure video-language training based on LLaVA series model. Proposed spatial-mid training with augmented spatial invariance to achieve SOTA result on zero-shot evaluation on Beacon3D benchmark.

VideoLLM for Physical Understanding. **07/2024 – 11/2024**
Brown University *Research Assistant*

- Collect and consolidated suites of physical reasoning benchmarks and datasets for VideoLLMs, including benchmark collection, task and integration of signals include depth and trajectory as physical priors.
- Constructed the evaluation and finetuning framework for VideoLLMs that probes failure modes in physical reasoning. Tested model, representation and prompting variants to improve attribute-level and scene-level physical understanding.

VEU-Bench: Towards Comprehensive Understanding of Video Editing. **07/2024 – 11/2024**
Opusclip *Research Intern*

- Introduced VEU-Bench, **the first comprehensive benchmark** for evaluating Video-LLM performance on 19 video editing understanding tasks, providing a thorough assessment of mainstream Video-LLMs.
- Proposed video editing understanding expert model Oscars, achieving a **28.3%** overall improvement on VEU-Bench and an average **3%** gain on general video benchmarks, surpassing existing SOTA models.
- Some more detail bullet points.

Temporal Sequence-Aware Model for Few-Shot Action Recognition. **03/2024 – 08/2024**
Zhejiang University *Hangzhou, China*

- Proposed a sequential perceiver-based temporal adapter that enables **temporal-aware video encoding** using the CLIP image encoder. Introduced the Unbalanced Optimal Transport Matching strategy for high-accuracy few-shot matching.
- Achieved **7.4%** accuracy improvement on HMDB51 over the previous SOTA, with an average **1.3%** gain across five major benchmarks, and a **4.1%** boost in cross-dataset evaluations, highlighting strong generalization of proposed framework.

Omni Representation Learning For Video CLIP **08/2023 – 12/2023**
Zhejiang University *Hangzhou, China*

- Introduced a Parallel Temporal Adapter (PTA) and a Self-Prompt Generator (SPG) module, and proposed a highly efficient spatial-temporal fusion mechanism for video perception across dynamic spatial-temporal scales.
- Achieved the best performance across four major video benchmarks with an average **1.5%** accuracy improvement. Surpassed the previous full-data method on the HMDB51 dataset by **1.5%** using only 16-shot training.

Professional Experience

Research Intern **4/2024 – 09/2024**
OpusClip *Palo Alto, CA*

- Built up online video clip quality judging system based on Gemini-pro and GPT-4o. Conduct comprehensive experiment on video judge dimensions. Raised **10%** in export rate and **8%** on clip like/dislike ratio.
- Developed instant video genre classification system based on Gemini. Enable fast input video sub-genre classification. Simplified four manual steps into one clip. Lifted **7%** improvement on credit consumption.
- Developed a screenplay-form video representation to improve long-term video understanding, achieving **68.8% accuracy** in breakpoint mode. Achieved **1st place** in CVPR 2024 Long-term Video Understanding Challenge.
- Proposed CoT-based sequential video agent system for long-term video comprehension. Achieved **45% accuracy** on hour-long video QA task and won **1st place** in the 2nd Perception Test Challenge's long-term video understanding track
- Proposed code-visual reward based reinforcement learning framework for VideoLLM on code-based dynamic chart generation, achived % performance gain with local 3B model and good trade-off between performance and efficiency as one of the motion graphic tool used in the Agent Opus video editing agent.
- Designed and built the prototype of Agent Opus framework, including context management, agent orchestration, prompt organization and playground.

Awards

- 1st place in the Hour-long videoQA challenge of the Second Perception Test Challenge @ECCV2024.
- 1st place in the Long-Term videoQA challenge of the LOVEU Workshop @CVPR2024.
- 2nd place in the Hour-long videoQA challenge of the Second Perception Test Challenge @ICCV2025.
- 3rd place in the Multi-View Foul Recognition challenge of the SoccerNet Challenge @CVPR2025.